Original Article

Structural Health Monitoring
1–14

DOI: 10.1177/14759217241249656
journals.sagepub.com/home/shm

Interpretable domain adaptation transformer: a transfer learning method for fault diagnosis of rotating machinery

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Abstract

Domain adaptation-based transfer learning methods have been widely investigated in fault diagnosis of rotating machinery, but their basic convolution or recurrent structure is subject to poor global feature representation ability, which hinders the learning of domain-irrelevant modulation information. In addition, the "black box" nature of deep learning models limits their applications in high risk-sensitive scenarios. In this paper, an interpretable domain adaptation transformer (IDAT) is proposed for the transferable fault diagnosis of rotating machinery. First, a multi-layer domain adaptation transformer framework is proposed, which can capture the global information that is crucial for learning the modulation information of different domains, and meanwhile reduce the feature distribution discrepancy. Second, an ensemble attention weight is applied to enable the transfer learning framework to be interpretable, which is implemented by averaging the integral values of the multi-head attention maps along the key direction. In addition, the raw vibration signals are embedded as the input of the model, which provides an end-to-end fault diagnosis. The proposed IDAT is tested by various cross-condition and cross-machine bearing fault diagnosis tasks, and results confirm the advantages of the method.

Keywords

Domain adaptation transformer, rotating machinery, interpretable, fault diagnosis

Introduction

Rotating machinery is widely used in manufacturing systems. However, their key components, such as rolling bearings, are vulnerable to various faults due to the long-time operation and harsh environments, such as high speeds, high temperature, and heavy load.[1,2] A simple fault may cause the breakdown of the entire system, and thus economic loss and even human casualties.[3] Therefore, the fault diagnosis of rotating machinery is crucial for ensuring the safe operation and reducing the maintenance cost of manufacturing systems.

In recent years, with the rapid development of deep learning (DL) techniques and hardware computing ability, DL-based fault diagnosis has achieved ever-increasing attention.[4–6] Different from shallow machine learning methods, such as support vector machine (SVM),[7] self-organizing map (SOM)[8] and random forest,[9] DL methods can mine the features automatically from the raw data by constructing multiple layers and thus overcome the dependence on manual design features that are necessary in traditional methods. Many typical methods, such as deep belief networks (DBN), long short-term memory networks and convolutional neural networks (CNNs), have been utilized in fault diagnosis.[10–12] For example, Shao et al.[13] employed compress sensing to reduce vibration data amount and then constructed a convolutional deep belief network for fault recognition. Liu et al.[14] applied Matrix profile

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to mine the signal segments containing fault information and then applied CNN to analyze these segments for bearing fault diagnosis. Chen et al.[15] combined CNN and cyclic spectrum coherence (CSCoh) for rolling bearing fault diagnosis, in which CNN was utilized to extract the features from the 2-D representations by CSCoh.

Although DL-based methods have been proven that can achieve high accuracy in fault diagnosis, their effectiveness depends on the basic assumption that the data used for training and test follow the same feature distribution.[16] In real applications, the different operation conditions and different machines lead to serious distribution discrepancy, resulting in low recognition accuracy, which makes DL methods poor generalization ability. Another limitation of DL methods in fault diagnosis is that they lack transparency in terms of their decisions due to the "black-box" nature, which becomes a major obstacle to some risk-sensitive scenarios.

Transfer learning (TL) provides a potential solution to address this issue, which leverages the knowledge learned from a source domain to facilitate the learning of a related target domain.[17–19] Domain adaptation is an effective tool to deal with the case that the data of source domain and target domain suffer from distribution discrepancy and there are no available labeled data in target domain.[20–22] Yang et al.[23] proposed a domain adaptation CNN for bearing fault diagnosis, which reduced the cross-domain feature distribution discrepancy. Li et al.[24] proposed a domain adaptation method via an adversarial training approach, which achieved promising results in the transfer tasks. These methods have made great contributions for fault diagnosis of rotating machinery. However, they exhibit poor ability in capturing global information due to the inherent limitations of basic DL structure,[25] which limits their performance in knowledge transfer. For example, without a transfer task, the features related to the amplitudes and shapes of fault impulses in a short time series may be enough to distinguish the health conditions, but for a transfer task, the fault information learned only from the short segments cannot guarantee a high accuracy because the fault impulses will be variable due to the variation in operation conditions or machines.[26]

Model explanation provides potential to reveal the decision mechanism, and it has gradually drawn researchers' attention. Ad hoc explanation and post hoc explanation are two major categories.[27,28] The former tends to guide the design of models for specific tasks,[29] but the generalization usually is poor. Post-hoc explanation is model-agnostic, while the decisions are inferred through additional modules or techniques.[30–32] These post-hoc interpretation methods have explained the decision to some extent. For example, in Grezmak et al.,[30] the vibration signals were converted into time–frequency images by wavelet transform, and CNN was used to classify different health conditions. Finally, layer-wise relevance propagation was used as an indicator to evaluate the contribution of different frequency bands to the diagnosis results. However, additional modules are necessary for explanation, resulting in high complexity. In addition, they usually combined with CNN and DBN, etc., and thus lack the global information capture ability.

Although the amplitudes or shapes of fault impulses change if they are collected from different operational conditions or machines, the amplitude modulation characteristics of signals from the same health condition are the same, which usually exhibit in a global view. Therefore, if a transfer model has a good global information capture ability, the transfer results will be improved accordingly. Fortunately, transformer has shown the great power in global information mining. It was proposed initially for natural language processing and computer vision, and has been gradually investigated in fault diagnosis. In addition, different from CNN and DBN, the inherent attention mechanism of transformer provides the potential for directly revealing the correlations between the components of input and their contribution on the decision.

The objective of the paper is to develop a TL model which could better capture the amplitude modulation by exploiting the unique merits of transformer in global information mining, and further explore the attention mechanism for model interpretation. To this end, an interpretable domain adaptation transformer (IDAT) is proposed to implement cross-condition and cross-machine fault diagnosis. First, the raw vibration signals are fed into an embedding module, and class token is performed to facilitate the learning of global information. Second, a multi-layer domain adaptation transformer is proposed to extract the useful features and reduce the feature distribution discrepancy. Then, an interpretation method is constructed based on the multi-head attention maps. Finally, the method is tested by experimental data, and results confirm the advantages of the method in cross-condition and cross-machine fault diagnosis tasks. The main contributions of the paper are as follows.

(1) A multi-layer domain adaptation transformer framework is proposed, which is expected to capture the modulation characteristics exhibited in a global view to facilitate knowledge transfer by fully exploiting the merits of transformer in global information mining. The raw vibration signals are embedded as input, and thus the model provides an end-to-end fault diagnosis.

(2) An ensemble attention weight is applied to enable the TL framework interpretable, which is implemented by averaging the integral values of the multi-head attention maps along the key direction. Compared to the multi-head attention maps, the result attention scores are more convenient to explain the decision, and are more accurate due to the integrating and averaging operations. In addition, because the interpretation is realized by the inherent attention mechanism of transformer, additional explanation modules are not involved.

(3) The proposed method is validated by experimental data, in which various cross-condition and cross-machine fault diagnosis tasks are conducted. The results present that IDAT performs best among comparison methods, and the ensemble attention weight enables the IDAT to be post-hoc interpretable.

Hereafter, this paper is organized as follows. In section "Methodology," we introduce the principle of transformer, and propose IDAT method. In section "Experimental evaluation," the proposed IDAT is validated by cross-condition and cross-machine tasks in rolling bearing fault diagnosis. Finally, section "Conclusion," draws the conclusions of the paper.

Methodology

Principle of transformer

Multi-head self-attention (MSA) is a key defining characteristic of transformer.[33,34] As shown in Figure 1, multiple attention relation calculations are conducted by MSA, which allows it to simultaneously deal with features of multiple positions. If the input is X, the input matrices of the MSA are formed as follows:

$$Q = XW^Q$$
$$K = XW^K$$
$$V = XW^V \tag{1}$$

where $Q \in R^{m\times n}$, $K \in R^{m\times n}$ and $V \in R^{m\times n}$ represent query, key and value, respectively; W^Q, W^K, and W^V denote weight matrices.

The self-attention is computed according to:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{n}}\right)V \tag{2}$$

where $\sqrt{n}$ is a scale factor. Parallel calculation of self-attention is conducted by

MSA, and then by calculating the transformation matrix W^o, the attention relations are transformed into a set of outputs

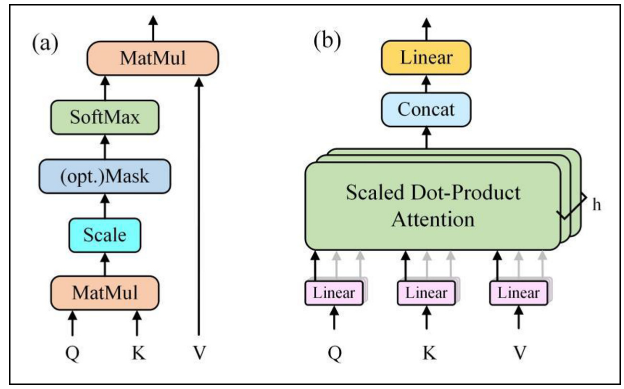


Figure 1. (a) Scaled Dot-Product attention, and (b) MSA. MSA: multi-head self-attention.

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \cdots, \text{head}_i)\text{W}^o \tag{3}$$

$$\text{head}_i = \text{Attention}\left(QW_i^Q, KW_i^K, VW_i^V\right) \tag{4}$$

The layer normalization is performed to avoid gradient explosion and vanishing, which is as follows[35]:

$$\text{LN}(x) = \frac{x-\mu}{\sqrt{\sigma^2+\varepsilon}} \odot g + b \tag{5}$$

where σ^2 and μ are variance and mean of x; g and b are the parameters scaling and translation; ε is generally set as 1e-5.

Feed-Forward neural network is conducted as nonlinear transformations and feature mapping as follows:

$$\text{FNN}(x) = (\text{ReLU}(x)W_1 + b_1)W_2 + b_2 \tag{6}$$

where $\text{ReLU}(x)$ is a rectified linear unit; W_1 and W_2 are weights; b_1 and b_2 are biases.

If x is the input and $f(x)$ is the weight layer output, the residual connection is calculated:

$$\text{RC}(x) = f(x) + x \tag{7}$$

Therefore, the complete forward propagation process of a transformer block is obtained as:

$$\tilde{x} = \text{LN}(\text{MSA}(x_i)) + x_i \tag{8}$$
$$x_o = \text{LN}(\text{FNN}(\tilde{x})) + \tilde{x}. \tag{9}$$

Interpretable domain adaptation transformer

To explore the merits of transformer model for TL and model interpretation in machine fault recognition, the IDAT is proposed. IDAT allows transferable feature learning from raw signals by designing a multi-layer domain shared transformer, and the construction of ensemble attention weight allows the interpretation of

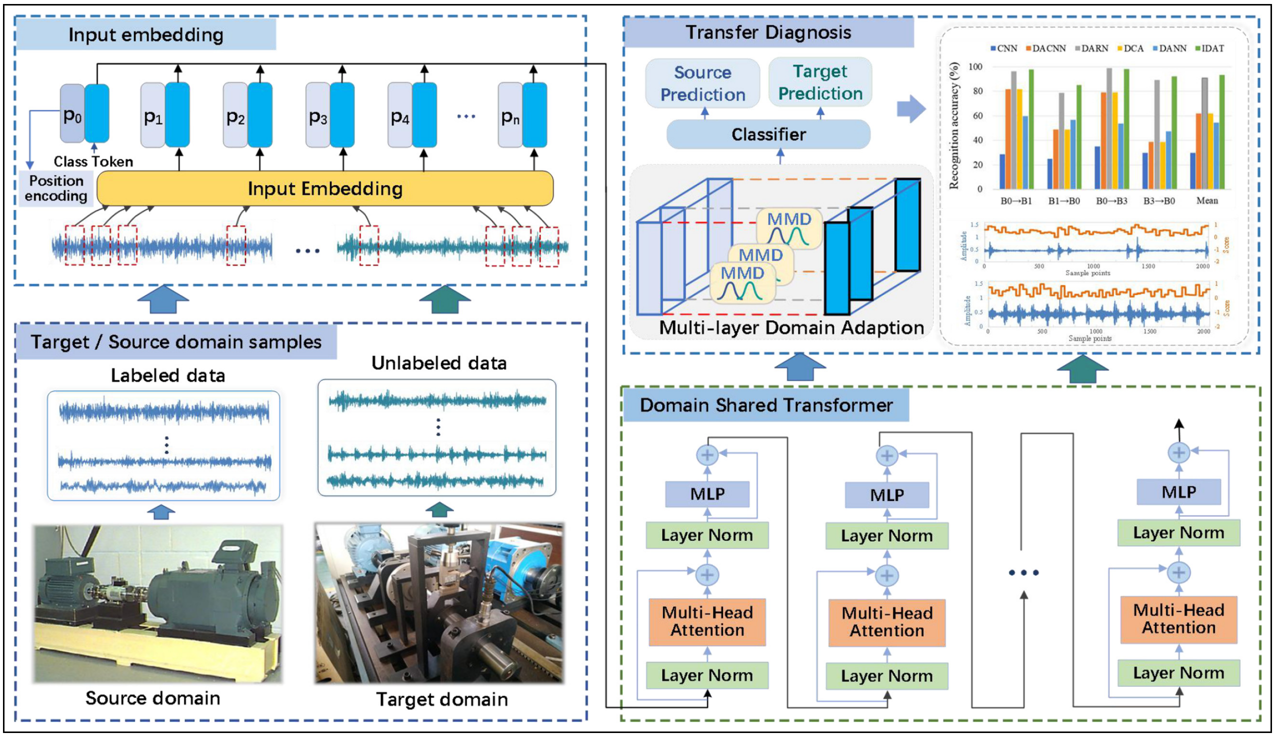


Figure 2. Flowchart of the proposed IDAT in fault diagnosis.
IDAT: interpretable domain adaptation transformer.

the data-driven "black box" DL model. The method is shown in Figure 2 and also is detailed as follows.

(1) Input of IDAT. To avoid any expert knowledge and achieve an end-to-end fault recognition, a raw signal embedding method is constructed. The input of the proposed model is composed of input embeddings, position encoding, and class token. The input embedding is a series of feature vectors obtained by applying a embedding module to raw vibration signals. The class token is a trainable feature vector that is formed and appended to the input embeddings, which facilitate the global information learning for enhancing the sensitivity of the model to input position. The position encoding is represented as sine and cosine functions as follows:

$$\mathrm{PE}(\mathrm{pos}, 2i) = \sin\left(\mathrm{pos}/10,000^{(2i/d_{\mathrm{model}})}\right) \tag{10}$$

$$\mathrm{PE}(\mathrm{pos}, 2i+1) = \cos\left(\mathrm{pos}/10,000^{(2i/d_{\mathrm{model}})}\right) \tag{11}$$

where pos represents the position, i denotes the dimension, and d_{model} is the dimension of hidden layer. It is seen that each dimension of position encoding is represented as a sine or cosine function with frequency $\omega_i = 1/10,000^{(2i/d_{\mathrm{model}})}$, and the waveforms constitute a geometric progression in the range of $2\pi - 10,000 * 2\pi$. Because $\mathrm{PE}_{\mathrm{pos}+k}$ can be obtained according to a linear function of $\mathrm{PE}_{\mathrm{pos}}$, the position encoding will allow the model to easily learn the relative positions. The input of IDAT is finally constructed as follows:

$$\begin{cases} X = \mathrm{Concat}(C, E) \\ Y = X + \mathrm{PE} \end{cases} \tag{12}$$

where C and E are the class token and input embeddings, and X is their combination; Y is the finial input.

(2) Feature representation and classification. The discriminative features are learned by performing non-linear mappings of multiple transformer layers, while the self-attention mechanism captures the dependencies between positions in the input sequence. This operation facilitates the learning of complex relationships and the mine of high-level features. The output of IDAT can be regarded as a function composed of both feature representation and attention weights.

$$F, A = \mathrm{Transformer}(Y) \tag{13}$$

where F are the learned features, and $A \in R^{H \times p \times p}$ are weights of attention. H and p are the numbers of heads and embeddings.

The obtained features in source domain will be further processed by a classifier which is implemented by a fully connected layer with softmax activation as:

$$\mathrm{softmax}(x_i) = \frac{e^{x_i}}{\sum_{j=0}^{p} e^{x_j}} \tag{14}$$

To minimize the classification error, the cross-entropy loss is applied to the features.

$$L = -{}^{1}/_{C} \sum_{i=1}^{C} (y_i * \log(p_i)) \tag{15}$$

where L is the cross-entropy loss, C is the number of classes, y_i is the true label, and p_i is the predicted probability distribution over the classes.

(3) Multi-layer domain adaptation. By minimizing the domain discrepancy, the transformer model can effectively leverage the diagnostic knowledge obtained from the source domain to improve the health condition recognition performance in the target domain. It learns to generalize and transfer the learned representations across domains, enabling it to make accurate health condition predictions on unseen target domain instances.

First, the discrepancy between the two domains is calculated by Maximum Mean Discrepancy (MMD)[36] which is a common and effective metric to measure the dissimilarity between probability distributions.

$$\mathrm{MMD}(X, Y) = \left\| (1/n_x) \sum_{i=1}^{n_x} \varphi(x_i) - (1/n_y) \sum_{j=1}^{n_y} \varphi(y_j) \right\|_H \tag{16}$$

where X and Y are the samples from two different domains, n_x and n_y are the sample numbers of each domain, $\varphi(x_i)$ and $\varphi(y_j)$ are the feature representations of the samples x_i and y_j in a high-dimensional Reproducing Kernel Hilbert Space (RKHS), and $||.||_H$ denotes the norm in the RKHS.

In this paper, the Gaussian kernel is applied to implicitly map the samples to the RKHS.[37] Let's denote the kernel function as $k\ (x,\ y)$ and $k(x,y) = \exp\left(-\|x-y\|^2 / 2\sigma^2\right)$, the MMD can be further approximated as:

$$\mathrm{MMD}\ (X, Y) = \left\| \sum_{i=1}^{n_x} \sum_{j=1}^{n_x} k(x_i, x_j)/(n_x^2) - 2/n_x n_y \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} k(x_i, y_j) + \sum_{i=1}^{n_y} \sum_{j=1}^{n_y} k(y_i, y_j)/\left(n_y^2\right) \right\|_H \tag{17}$$

After computing the MMD loss, the trainable parameters of the transformer model can be trained to minimize the domain discrepancy. This adjustment encourages the model to extract domain-invariant representations that capture shared diagnostic knowledge between the two domains.

To improve training efficiency, we have developed a joint training strategy to synchronously optimize health classification tasks and domain adaptation tasks. In addition, a hypermeter is further introduced to balance the cross-entropy (CE) loss and MMD, as shown in bellow:

$$\mathrm{Loss} = \mathrm{CE} + \lambda \mathrm{MMD} \tag{18}$$

This joint training strategy allows the health classification error CE loss and domain distribution difference MMD to be used together to supervise the learning of the model at each training step. Compared to independent optimization for each task, in joint training, data only needs one forward transfer to complete multi-task learning, improving training efficiency.

(4) Model interpretation. To enhance the interpretability and reliability of intelligent models, a novel ensemble attention weight calculation method is constructed based on the multi-head attention maps. This method enables the analysis of input importance in relation to diagnostic results, thereby providing a clear explanation of the network's internal reasoning mechanism. This method is detailed as bellow:

$$A_C = \frac{1}{h} \sum_{h=1}^{H} \sum_{i=1}^{1} A_{i,j}^h \tag{19}$$

where $A_{i,j}^h \in \mathbb{R}^{\mathbb{H} \times {}_{\mathrm{I}} \times {}_{\mathrm{I}}}$ represents the attention weight of i-th row and j-th column in h-th head, which refers to the similarity, measured by the scaled dot-product attention, between i-th embedding and j-th embedding. In addition, $A_C \in \mathbb{R}^{\mathrm{I}}$ represent importance factors, which can qualitatively evaluate the importance of the input to diagnosis. This is based on two aspects:

First, the importance factors correspond to the average attention weights of the first row in H heads. The first row of the attention map captures the correlation between the class token and all input embeddings, indicating their relative significance in the attention mechanism.

Second, the class token is considered as the final representation for the entire input, offering a comprehensive summary of the input vibration signal. It serves as an overview or global context representation, encapsulating the most relevant information from the entire sequence, which aids in capturing the essential characteristics and patterns for the given task or classification.

In addition, the obtained importance factors are linearly scaled to the range from zero to one. It is transformed by the formula:

$$y = \frac{x - \min(x)}{\max(x) - \min(x)} \tag{20}$$

where x is the input vector, y is the linear scaled vector, $\min(\bullet)$ refers to the minimum of the input vector, and $\max(\bullet)$ means the maximum of the input vector.

Experimental evaluation

Description of datasets

Two datasets that collected from rolling bearings are adopted to evaluate the method. The parameters of the tested bearings are listed in Table 1.

The first dataset is collected from our lab at Beijing University of Technology, which is called "BJUT dataset" in this paper. The experimental setup is shown in Figure 3, which consists a motor, a toque, a load device and the tested rolling bearing. The accelerometer is used to collect signals, and the sampling rate is set as 12 kHz. Four health conditions, that is, normal condition (NC), outer race fault (OF), inner ring fault (IF) and roller fault (RF) as listed in Figure 4, are considered. There are four operation conditions, marked as B0, B1, B2 and B3, as shown in Table 2, which is divided according to the different radial forces and the rotating speeds. Each operation condition contains 800 samples, and each sample owns 2048 sample points.

The second dataset is the most popular data collected from Case Western Reserve University, which is marked by "CWRU dataset." The data of the drive end bearings with different health conditions, that is, NC, OF, IF, and RF, were collected by the accelerometer. To better illustrate the location of the tested bearings, one diagram of the test rig is provided in Figure 5. The sampling rate is set as 12 kHz. The dataset is also grouped into four operation conditions, labeled as C0, C1, C2, and C3, presented in Table 3. Each operation condition contains 800 samples, and each sample owns 2048 sample points.

Because the dataset of CWRU is widely analyzed in intelligent fault diagnosis, we analyze the vibration characteristics of our BJUT dataset. Figure 6 shows one sample randomly selected form each health condition (i.e., NC, OF, IF, and RF) under each operation condition (i.e., B0, B1, B2, and B3). It can be seen that there are no obvious fault impulses in the waveforms of NC. For OF, there are various fault impulses in the waveforms of all operation conditions, and these impulses own nearly the same amplitudes and appear with equal intervals. Different from OF, the amplitudes of fault impulses of IF change due to the variation in load of the defect location. For RF, the fault impulses are not as clear as those of OF and IF. Because the fault impulses of RF are weak, it seems hard to distinguish NC and RF such as the two samples under operation condition B2. Further analysis finds that when the radial force is small (B0 and B3), the waveforms of IF contain more noise which make them similar to that of RF. Therefore, it is hard to identify the IF and RF based on the shapes of waveforms or periodicity characteristics of fault impulses.

The common features, that is, root mean square (RMS), peak-to-peak (P-P) and kurtosis, of all samples under operation conditions B0 and B2 are calculated and presented in Figure 7. It is seen that the RMS and P-P values of some samples of RF are mixed with that of NC and OF, which means the RMS and P-P values cannot classify the health conditions effectively. For kurtosis values, the distribution range of RF covers all the other three health conditions. The above analysis confirms that the common features cannot be directly applied to classify the health conditions of rolling bearings. In addition, because the vibration signals collected from different machines exhibit serious distribution discrepancy, if the data of target machine lacks labels, it is impossible to recognize the health conditions via the

Table 1. Parameters of the tested rolling bearings.

Datasets	Type	Outer race diameter (mm)	Inner race diameter (mm)	Ball race diameter (mm)	Ball number	Contact angle
BJUT	LUT6010	80.00	60.00	8.75	13	0
CWRU	6205-2RS	52.00	25.00	7.94	9	0

BJUT: Beijing University of Technology; CWRU: Case Western Reserve University.

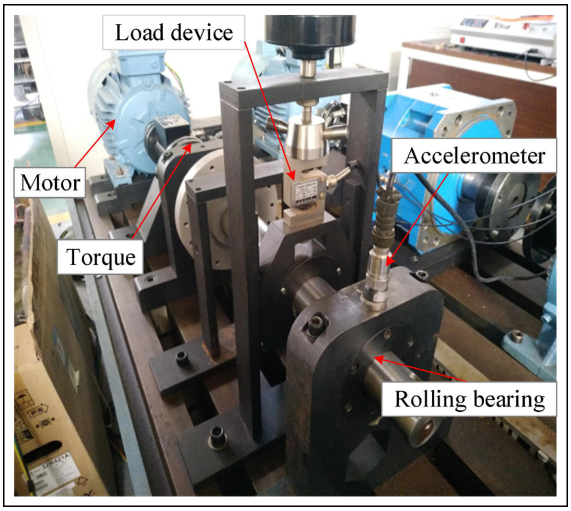


Figure 3. Experimental setup of BJUT dataset.
BJUT: Beijing University of Technology.

Table 2. Details of BJUT dataset.

BJUT	Health conditions	Radial force (kN)	Rotating speed (rpm)
B0	4	1.1	1800
B1	4	0	1800
B2	4	1.1	1500
B3	4	0	1500

BJUT: Beijing University of Technology.

Table 3. Details of CWRU dataset.

CWRU	Health conditions	Motor load (HP)	Rotating speed (rpm)
C0	4	0	1797
C1	4	1	1772
C2	4	2	1750
C3	4	3	1730

CWRU: Case Western Reserve University.

model optimized using the data collected from another machine.

Cross-condition fault diagnosis

In this section, the proposed IDAT is evaluated in cross-condition fault diagnosis. The parameter of kernel size σ in Equation (18) and the tradeoff parameter λ in Equation (19) play a crucial role in the transfer results, and thus they are analyzed firstly. The experiment is conducted on task B0 $\rightarrow$ B1, that is, the data B0 is used for training and the data B1 is used for test. The kernel size σ is selected from $\{4, 8, 16, 32\}$, and the parameter λ is searched from $\{0.125, 0.25, 0.5, 1, 2\}$. Each experiment is conducted five times and the average accuracies are presented in Figure 8. It is seen that when σ and λ are set as 16 and 0.5, respectively, the accuracy is the highest. To avoid random interference, the other transfer tasks in the cross-condition fault diagnosis are listed in Figure 9, and the tasks in section "Cross-machine fault diagnosis" share the same parameters.

Four cross-condition fault diagnosis tasks (B0 $\rightarrow$ B1, B1 $\rightarrow$ B0, B0 $\rightarrow$ B3, and B3 $\rightarrow$ B0) are conducted, and each task is performed for five times to avoid randomness. The results of the proposed IDAT and three comparison methods, that is, CNN, domain adaptation CNN (DACNN),[38] domain adaptation ResNet (DARN),[39] deep correlation alignment (DCA),[40] and domain adversarial neural network (DANN),[41] are provided in Figure 9. The standard deviation values of all methods for all tasks are presented in Figure 10. The proposed IDAT achieves the highest average accuracy of 96.90% with the lowest average standard deviation value 0.63% among these methods. For all tasks, the accuracy of IDAT is also the highest, which confirms the advantage of IDAT in cross-condition fault diagnosis. It can also be found that, without domain adaptation operation, the CNN only obtains the accuracy of 57.70%, which means TL will help to improve the recognition results. It should

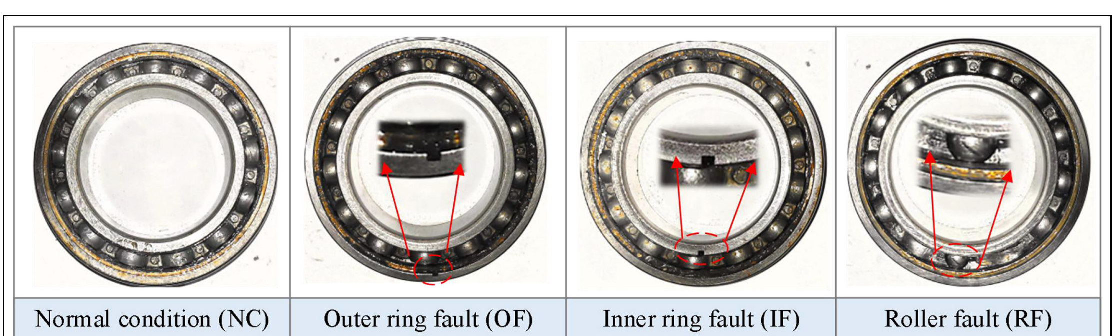


Figure 4. Rolling bearings used in BJUT dataset.
BJUT: Beijing University of Technology.

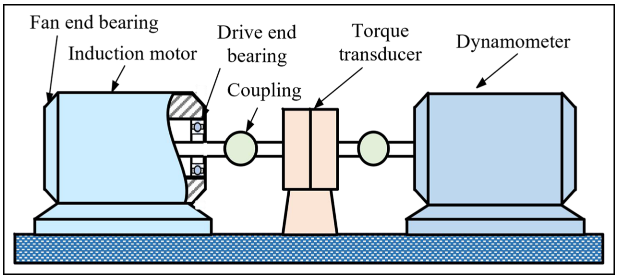


Figure 5. Diagram of drive end bearing test rig.

be noted that although for tasks B0 → B1 and B0 → B3, the standard deviations values of CNN are lower than that obtained by IDAT, the corresponding accuracies are only 50.30% and 50.20%, and the average value for the four tasks of CNN is 6.38%, which is much higher than 0.63% obtained by IDAT. The above analysis confirms the advantages of IDAT in cross-condition fault diagnosis.

Cross-machine fault diagnosis

In this section, the proposed IDAT is evaluated and compared in the cross-machine fault diagnosis. Four tasks are performed, and each task is conducted five times. It should be noted that the parameters σ and λ are set as the same as those in section "Description of datasets". The recognition accuracies and the standard deviations are also provided in Figures 11 and 12, respectively. It is seen that, without domain adaptation operation, the mean recognition rate of CNN is only 29.63%, which is near to the randomly classification rate 25%. The accuracy of DARN is up to 90.72%, respectively, but they are still lower than 93.34% obtained by IDAT. The proposed IDAT also obtains the lowest mean standard deviation. This further confirms the advantages in cross-machine fault diagnosis.

The confusion matric of different methods are also compared, which are from the first computation in task B0 → C0. As presented in Figure 13, due to the distribution discrepancy across domains, a great number of samples are misclassified by CNN. For example, the samples of NC are all misclassified as RF. In contrast, DACNN, DARN, DCA, and DANN achieve better results than CNN through TL. For the IDAT, there are 5.2% samples in RF that are misclassified as IF, and the other three health conditions are all recognized accurately. It is more difficult to distinguish the IF and RF theoretically because they own similar amplitude modulation characteristics in the time-domain signals due to the load variation during the bearing rotation.

To visually compare the proposed IDAT with the other methods, the t-stochastic neighbor embedding (t-SNE) is adopted to analyze the extracted features. Figure 14 (a) to (d) show the results by CNN, DACNN, DARN, DCA, DANN, and proposed IDAT, respectively. It is seen that the learned features corresponding to the same health conditions of different machines obtained by CNN are subject to serious distribution discrepancy. For example, the learned features of NC of the two machines are separated well, while the ideal result is that they are clustered as a

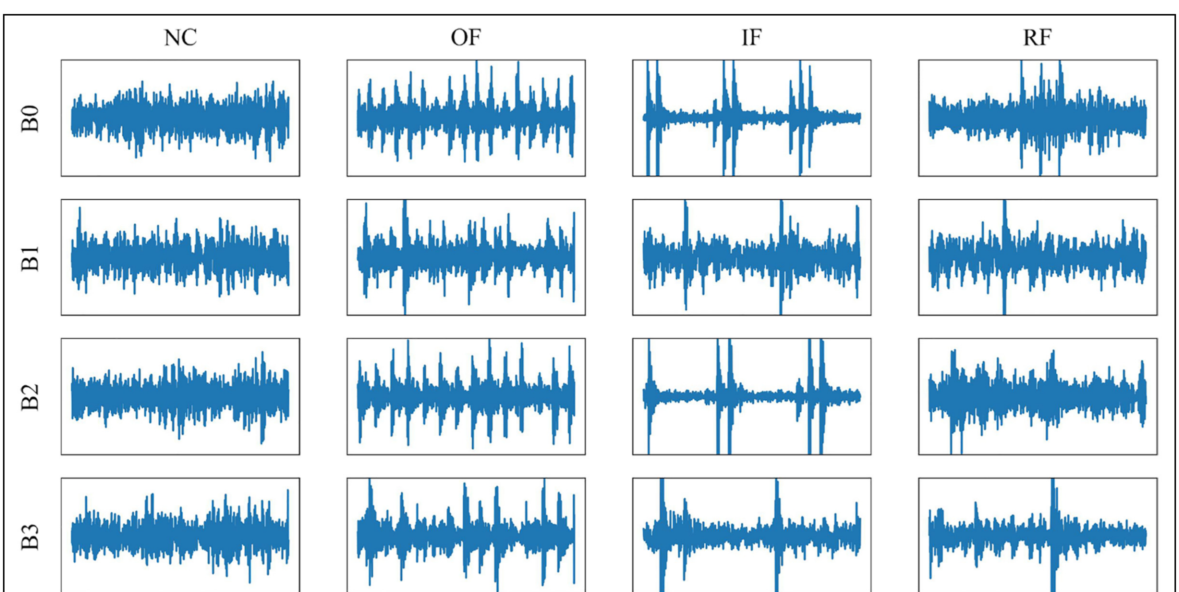


Figure 6. Waveforms of the BJUT dataset.
BJUT: Beijing University of Technology.

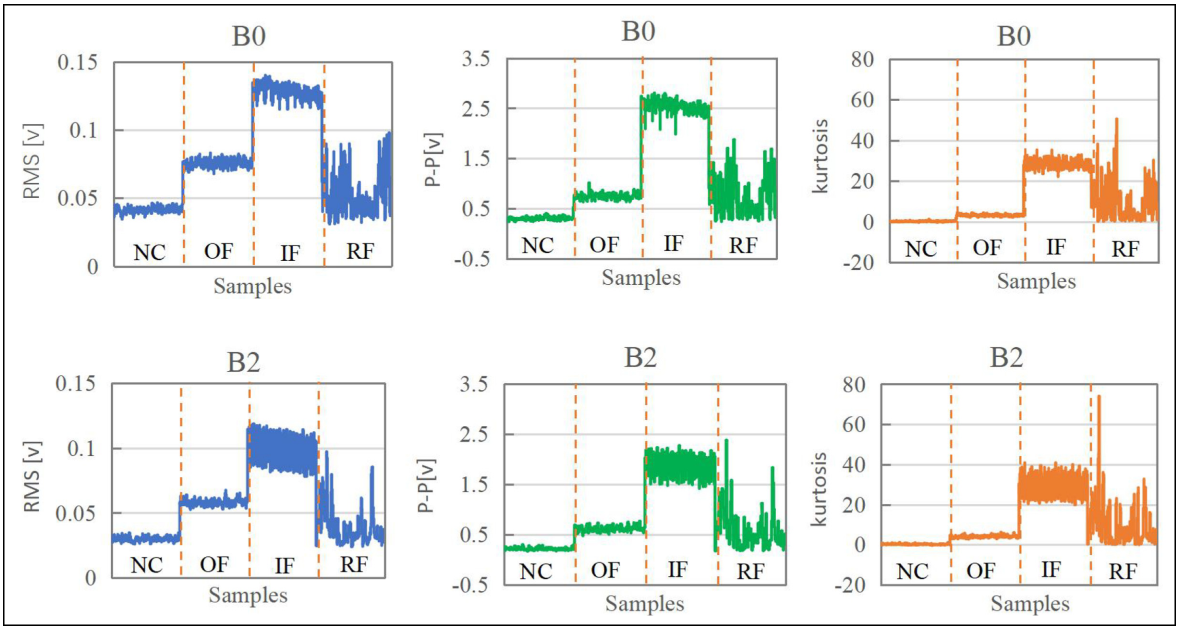


Figure 7. RMS, P-P, and kurtosis values of BJUT dataset.
BJUT: Beijing University of Technology; RMS: root mean square; P-P: peak-to-peak.

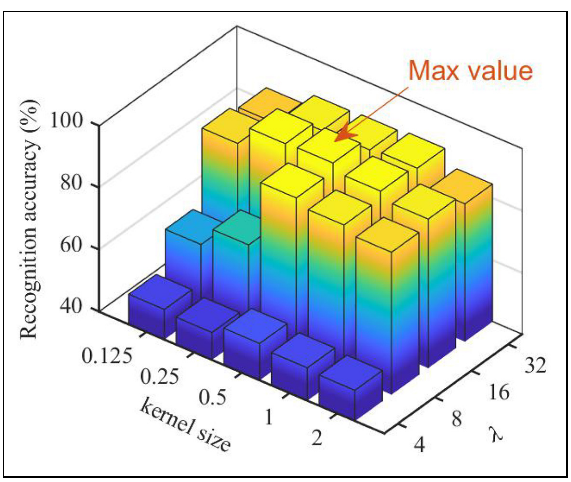
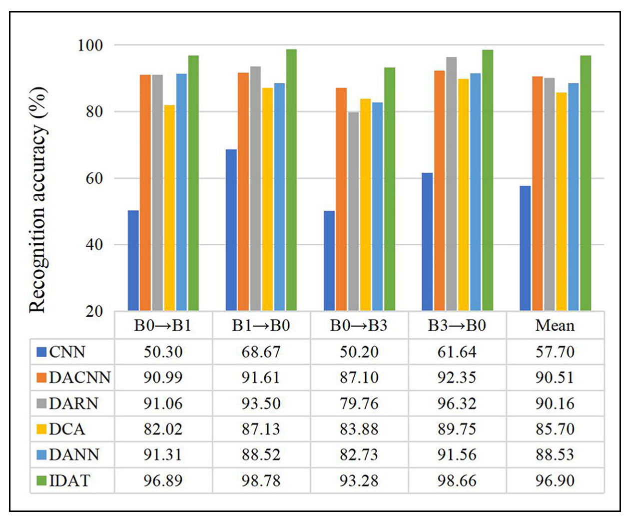


Figure 8. Effect of kernel size σ and λ on the transfer results (100 epochs are conducted).



	B0→B1	B1→B0	B0→B3	B3→B0	Mean
CNN	50.30	68.67	50.20	61.64	57.70
DACNN	90.99	91.61	87.10	92.35	90.51
DARN	91.06	93.50	79.76	96.32	90.16
DCA	82.02	87.13	83.88	89.75	85.70
DANN	91.31	88.52	82.73	91.56	88.53
IDAT	96.89	98.78	93.28	98.66	96.90

Figure 9. Comparisons of accuracies in cross-condition fault diagnosis.

class. By TL, the distribution discrepancy of features corresponding to the same health conditions from the two machines is corrected to some degree, but the results of DACNN, DARN, DCA, and DANN are still unsatisfied. For DACNN, some samples in IF of target domain are clustered into the OF, while for DARN, the clusters corresponding to IF and RF are very close and even mixed. The above analysis confirms the advantages of the proposed IDAT in cross-machine fault diagnosis.

To explain the learned knowledge of different layers, the attention maps of MSA mechanism are visualized as shown in Figure 15. Figure 15(a) and (b) show the results of OF and IF in source domain, while Figure 15(c) and (d) show that in target domain. There are totally four attention maps in each layer, and the layer levels increase from top to bottom with the vertical axis. It is seen that the model almost pays attention evenly to each location on the attention maps for the first two layers, but more attention are paid to some local locations as the layers increase. In addition, high weights are distributed as lines vertical to horizontal axis K, which means more attention are paid to some

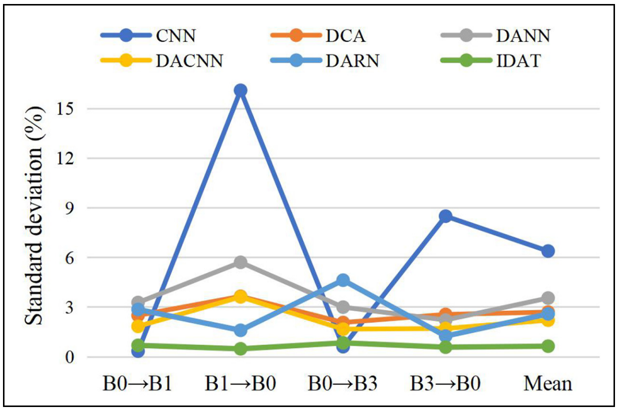


Figure 10. Comparisons of standard deviations in cross-condition fault diagnosis.

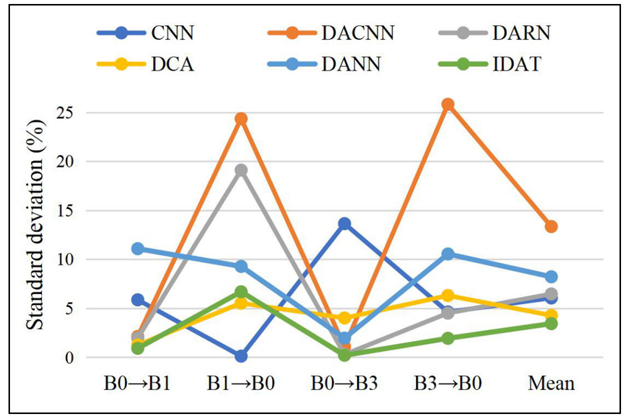
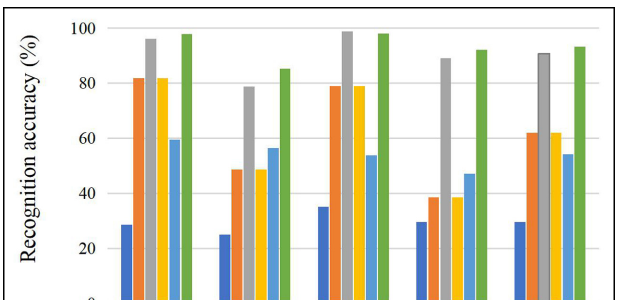


Figure 12. Comparisons of standard deviations in cross-machine fault diagnosis.



	B0→B1	B1→B0	B0→B3	B3→B0	Mean
CNN	28.70	25.03	35.16	29.64	29.63
DACNN	81.87	48.67	79.06	38.55	62.04
DARN	96.16	78.79	98.76	89.15	90.72
DCA	81.87	48.67	79.06	38.55	62.04
DANN	59.57	56.56	53.77	47.26	54.29
IDAT	97.91	85.23	98.07	92.14	93.34

Figure 11. Comparisons of accuracies in cross-machine fault diagnosis.

specific signal segments of the input. Generally, no matter the source domain or the target domain, the bright lines in the attention maps for OF are denser compared that for IF. This can be explained by the fact that the amplitudes of fault impulses caused by OF are more even, but for IF, they change with time due to the amplitude modulation.

To illustrate the model more intuitively, the attention scores by the proposed ensemble attention weight method are demonstrated along with the raw signals of different health conditions as shown in Figure 16. Figure 16(a) to (d) correspond to NC, OF, IF, and RF, respectively. For NC, there are no obvious impulses in the waveform, and the waveform mainly contains noises and the inevitable low-frequency fluctuation caused by errors of manufacturing and installation. Correspondingly, the attention scores are earned evenly. For OF, the impulses appear evenly because the load in the fault location is constant with rotating, and correspondingly, the larger scores are earned by these impulses. For IF and RF, the amplitudes of impulses are not equal due to the effect of amplitude modulation caused by load variation. As Figure 17 shown, when the fault is at the bottom of the bearing, the load is the largest and then the highest amplitude. The locations of the defects of inner race and rolling elements change following with the rotating of shaft which is fixed on the inner race. Therefore, the loads at the defects of these two parts also change periodically with the rotating frequency of support shaft. For better illustration, one black dotted line is, respectively, plotted in Figure 16(c) and (d) to present the amplitude modulation caused by load variation. From Figure 16(c), it is seen that when the fault impulses own higher amplitudes, the larger attention scores are earned. For the RF, in addition to the revolution of rollers around the shaft, they also rotate around their own axes, so their movement is more complex, which combined with the effect of noise and the vibration of other components makes their fault impulses not prominent (which can be found from Figure 6, and also be certified by the fluctuations of the features obtained from RF shown in Figure 7). Even though the fault impulses are weak and the modulation characteristics are more complex, the attention scores present a rough similar trend with amplitude modulation. For comparison, the activation map of DACNN is also analyzed as shown in Figure 18. It can be found that even for the simple IF, the activation map cannot reflect the amplitude modulation because there are many higher activation values at the locations with lower amplitudes. For RF, the performance of DACNN is also worse than that obtained via the proposed method. The above analysis confirms that what are captured by the IDAT can be explained.

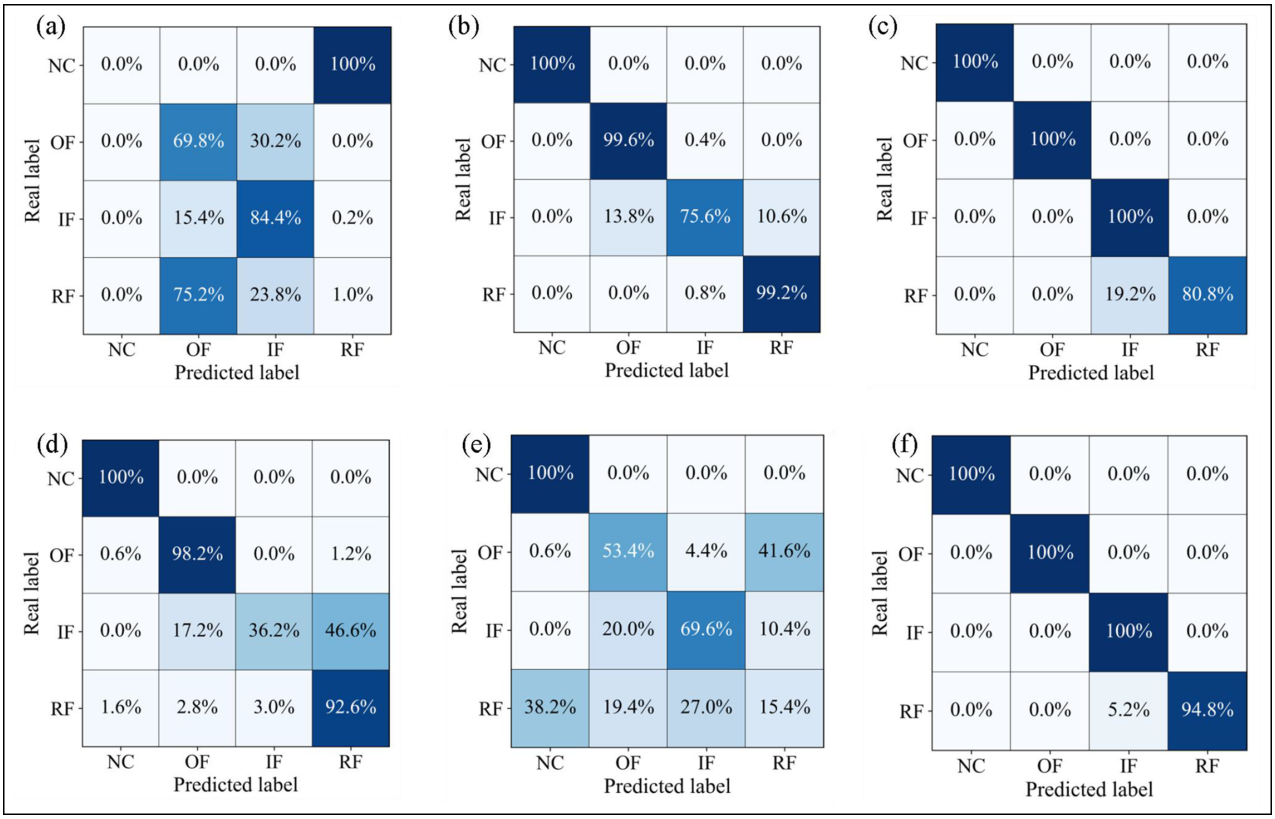


Figure 13. Confusion matrix obtained by (a) CNN, (b) DACNN, (c) DARN, (d) DCA, (e) DANN, and (f) IDAT.
CNN: convolutional neural network; DACNN: domain adaptation CNN; DARN: domain adaptation ResNet; DANN: domain adversarial neural network; IDAT: interpretable domain adaptation transformer.

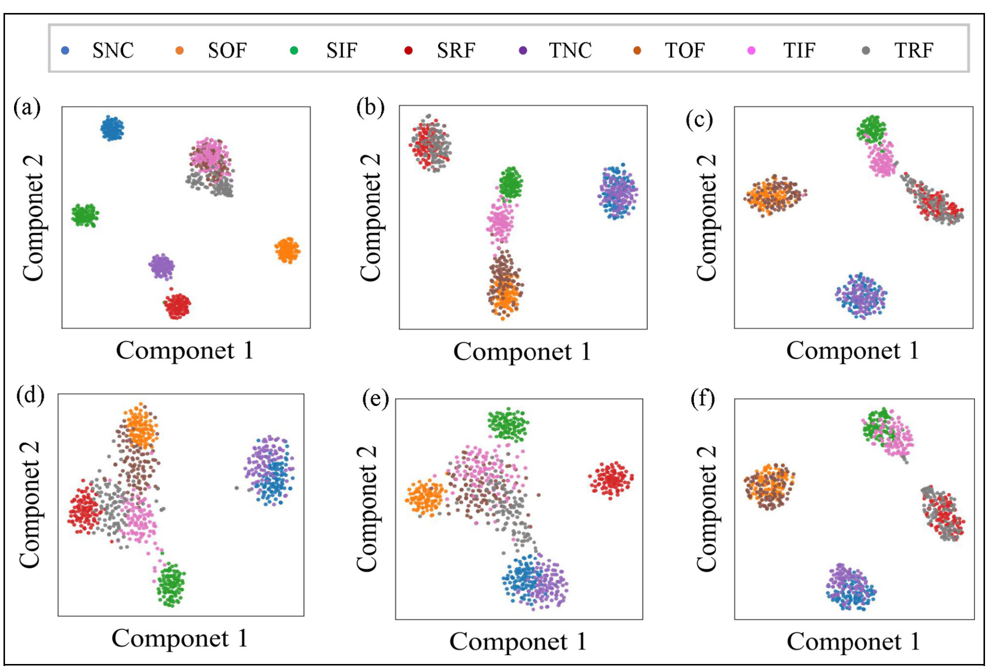


Figure 14. Visualization results of learned features by (a) CNN, (b) DACNN, (c) DARN, (d) DCA, (e) DANN, and (f) IDAT ("S" and "T" are added to labels to present source domain and target domain).
CNN: convolutional neural network; DACNN: domain adaptation CNN; DARN: domain adaptation ResNet; DANN: domain adversarial neural network; IDAT: interpretable domain adaptation transformer.

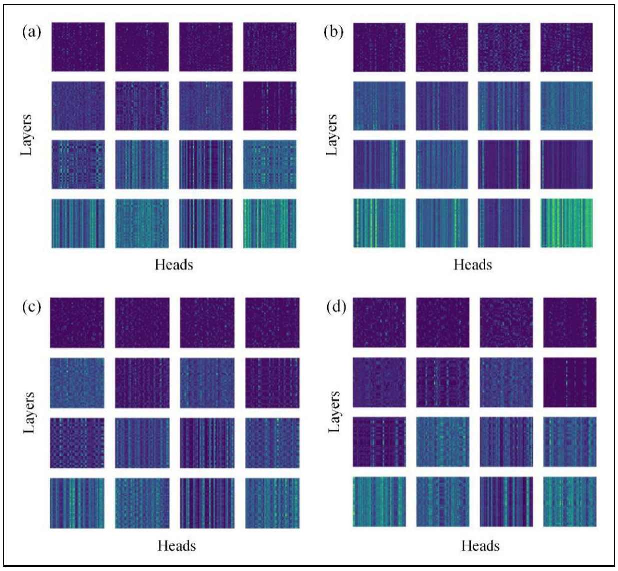


Figure 15. Visualization of attention maps of the health conditions: (a) and (b) correspond the OF and IF in source domain; (c) and (d) correspond the OF and IF in target domain.
OF: outer race fault; IF: inner ring fault.

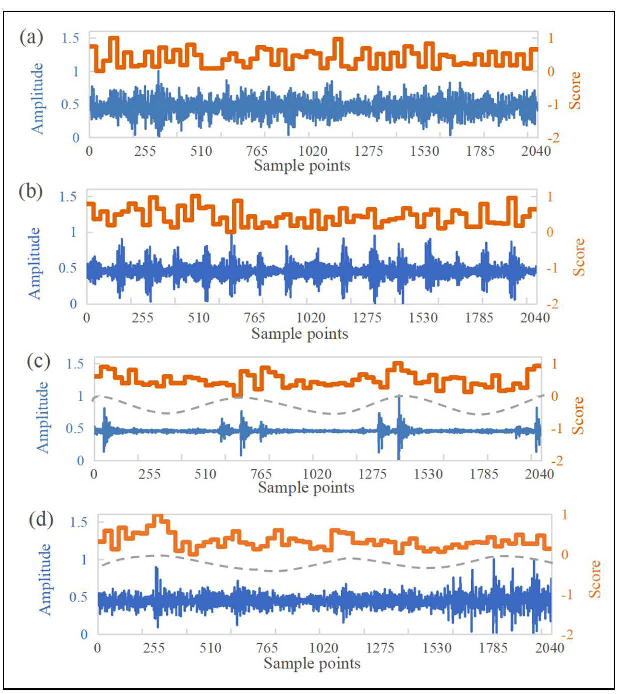


Figure 16. Waveforms and attention scores obtained by the proposed method of health conditions (a) NC, (b) OF, (c) IF, and (d) RF.
NC: normal condition; OF: outer race fault; IF: inner ring fault; RF: roller fault.

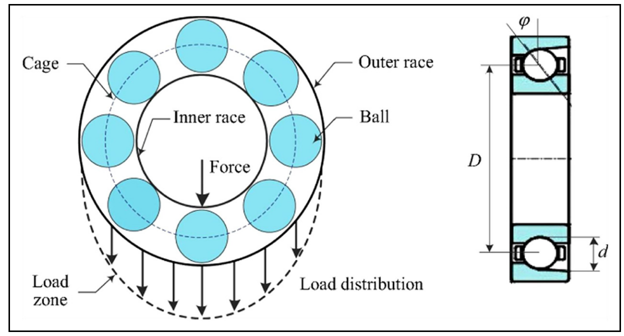


Figure 17. Illustration of load zone of rolling bearings.[8]

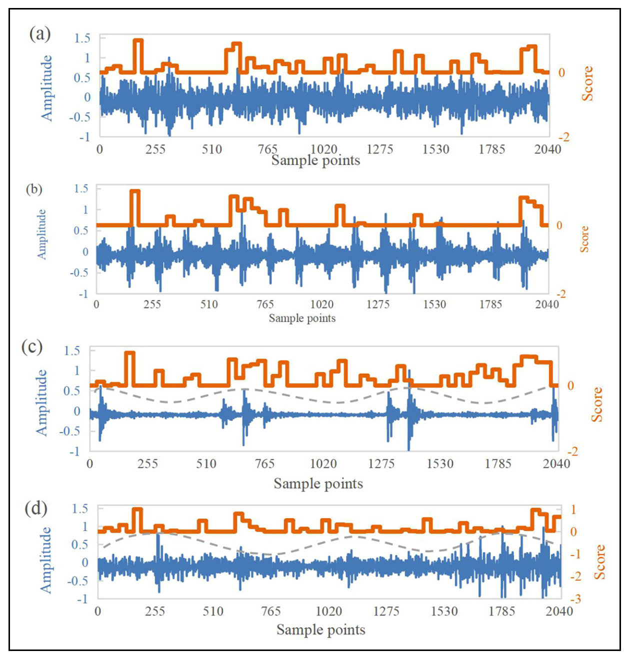


Figure 18. Waveforms and activation maps obtained by DACNN of health conditions (a) NC, (b) OF, (c) IF, and (d) RF.
CNN: convolutional neural network; DACNN: domain adaptation CNN; NC: normal condition; OF: outer race fault; IF: inner ring fault; RF: roller fault.

Conclusion

In this paper, an IDAT is proposed for the knowledge transfer in rotating machinery fault diagnosis. A multi-layer domain adaptation transformer framework is proposed, which can not only reduce the feature distribution discrepancy but also capture the global information that is beneficial to mine the amplitude modulation information of the vibration signals in different domains. An ensemble attention weight is constructed to enable the TL framework interpretable. An input embedding module is constructed and thus provides an end-to-end fault diagnosis. The proposed IDAT is tested and compared with several methods in various cross-condition and cross-machine fault diagnosis tasks, and results confirm the advantages of the method.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This study is supported by National Natural Science Foundation of China (52305086 and 52075008) and Beijing Municipal Natural Science Foundation (L221007).

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